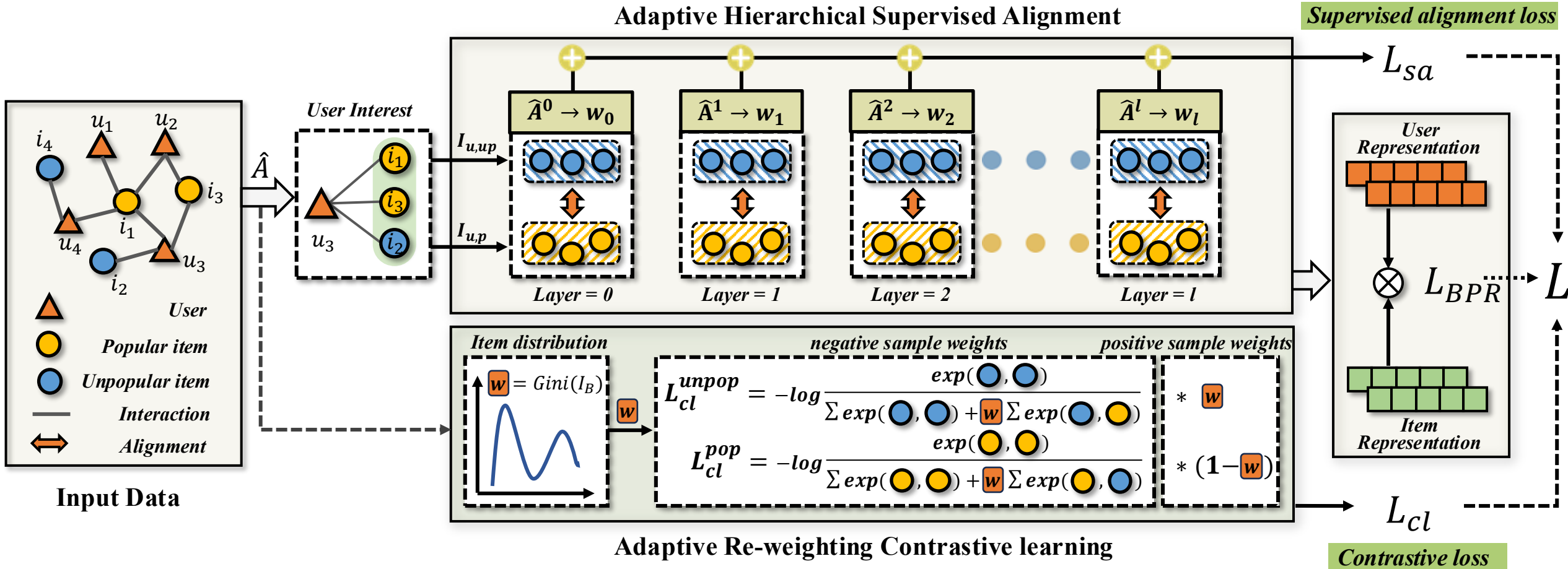




$I_{u,(u)p} : (Un)popular \text{ item interacted by user } u$

$\hat{A} : \text{normalized adjacency matrix}$

$I_B : \text{items in a mini-batch}$

$L : \text{total loss}$